



Problem Title: Reference Object Recognition Under Viewpoint and Angle Variations

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Team Details:

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The Challenge: Accurately locating metallic industrial parts in a manufacturing environment is difficult due to:

- **Specular Reflections (Glare):** Shiny surfaces hide texture, confusing standard vision sensors.
- **Scale Variations:** The camera distance changes (zoom), making the object appear up to 2.5x larger or smaller.
- **Perspective Tilt:** Objects are rarely perfectly flat; camera angles cause distortion.



Solution Overview: A robust Desktop Application that uses Computer Vision with a novel "Adaptive Trust" logic.

- **Glare-Resistant:** Uses CLAHE (Contrast Limited Adaptive Histogram Equalization) to "see" texture on shiny metal.
- **Geometric Integrity:** Forces Rigid Affine transformation to ensure the bounding box remains a perfect rectangle (no skewing).
- **Adaptive Scale Tolerance:** Dynamically adjusts validation rules.
- **Strong Matches (6+ points):** Allowed to zoom/scale widely (0.5x - 3.0x).
- **Weak Matches (<6 points):** Strictly clamped (0.8x - 1.2x) to prevent false positives.

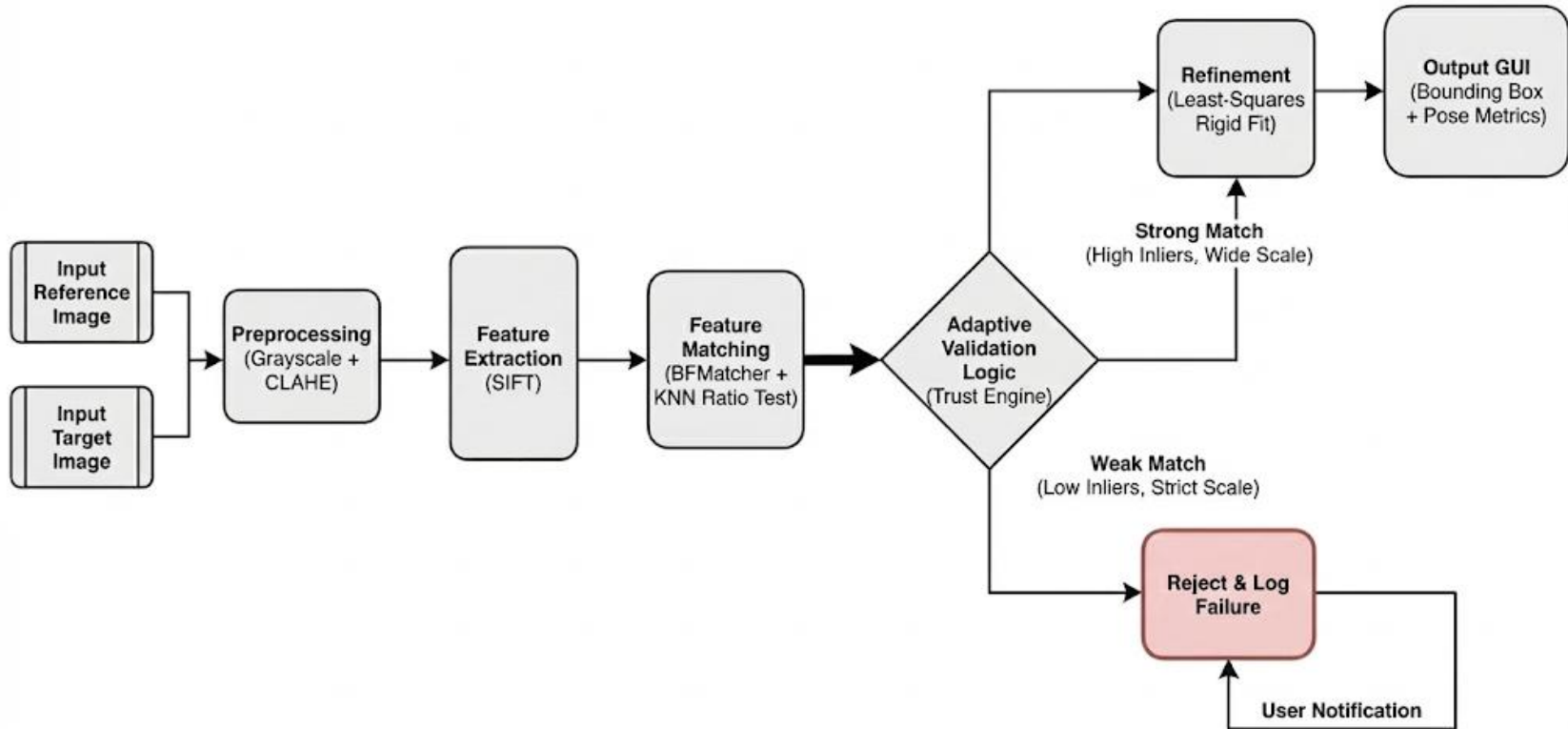


- **Programming Language:** Python 3.x
- **Computer Vision Library:** OpenCV (cv2)
- **Algorithms:** SIFT, CLAHE, AffinePartial2D, RANSAC

1. Numerical Processing: NumPyUser

- **Interface (GUI):** PyQt5 (Qt for Python)
- **Features:** Real-time logging, interactive ROI drawing, visual overlays.







Input Data: Loads Reference and Target images.

Preprocessing: Applies Grayscale conversion and CLAHE (Contrast Limited Adaptive Histogram Equalization) to remove metal glare.

Feature Pipeline:

Detection: SIFT (Scale-Invariant Feature Transform) extracts keypoints.

Matching: Brute-Force Matcher with K-Nearest Neighbors ratio test (0.85) filters noise.

Geometry Engine: Uses RANSAC (Random Sample Consensus) with Rigid Affine Constraints to prevent distortion.

Validation Logic: Adaptive Trust Engine dynamically validates matches based on Inlier Count vs. Scale.

Output: Renders precise Bounding Box and Pose Metrics (X, Y, Rotation, Scale)



Preprocessing: CLAHE Enhancement

- **Technique:** Applied CLAHE (Contrast Limited Adaptive Histogram Equalization) with ClipLimit = 4.0.
- **Why:** Superior to global equalization. It maximizes local contrast to reveal keypoints on specular metallic surfaces.

Geometric Solver: Rigid vs. Homography

- **Algorithm:** Replaced find Homography (8 DoF) with estimateAffinePartial2D (4 DoF).
- **Benefit:** Constrains the transformation to Translation + Rotation + Scale. This mathematically prevents "Shearing" artifacts (the "Flying Box" error).

Validation Logic: Adaptive RANSAC

- **Logic:** Implemented a dynamic thresholding engine based on inlier density:
- **High Inliers (≥ 6):** Relaxes Scale constraints (0.5x-3.0x) to handle perspective zoom.
- **Low Inliers (< 6):** Tightens Scale constraints (0.8x-1.2x) to reject false positives.

Optimization: Sub-Pixel Refinement

- **Method:** Performed a final Least-Squares fit on the consensus inlier set.
- **Result:** Minimizes re-projection error, achieving sub-pixel bounding box accuracy.



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Results Achieved

- Reliable under zoom, tilt, and glare.
- Stable after fixing the skewed box using Rigid Affine math.
- Smart noise vs object filtering.

Performance

- Speed: < 200 ms per frame.
- Accuracy: Sub-pixel precision.

Key Insights

- Math > Generic AI for precision.
- Human logic: Trust clear views, doubt blurry ones.

Limitations

- Fails if object is heavily blocked (<3 points).
- Motion blur reduces accuracy.
- SIFT is accurate but computationally heavy.

Next Steps

- Connect to RTSP/USB cameras.
- Add object memory/tracking.
- Use a small AI for faster initial detection.